

Relative-Bregman uniqueness for the L^1 -gradient flow and the full strongly elliptic quadratic subcase

Codex agent lane 8 (GPT5.6)

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Status. Candidate partial result, likely valid, human review requested. The result does not settle uniqueness for every weak minimizing-movement limit of a general strongly convex integrand.

1 Source question

Chambolle and Novaga study the L^1 -minimizing movement for

$$\Phi(u) = \int_{\Omega} F(Du)$$

on a bounded connected domain, with homogeneous Neumann or Dirichlet boundary conditions. They obtain limit solutions, characterize them for smooth strongly convex F , prove uniqueness for monotone solutions, and prove unconditional uniqueness for the isotropic Dirichlet energy $F(\xi) = |\xi|^2/2$ [1]. Their abstract states:

L^1 -GRADIENT FLOW OF CONVEX FUNCTIONALS

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ABSTRACT. We are interested in the gradient flow of a general first order convex functional with respect to the L^1 -topology. By means of an implicit minimization scheme, we show existence of a global limit solution, which satisfies an energy-dissipation estimate, and solves a non-linear and non-local gradient flow equation, under the assumption of strong convexity of the energy. Under a monotonicity assumption we can also prove uniqueness of the limit solution, even though this remains an open question in full generality. We also consider a geometric evolution corresponding to the L^1 -gradient flow of the anisotropic perimeter. When the initial set is convex, we show that the limit solution is monotone for the inclusion, convex and unique until it reaches the Cheeger set of the initial datum. Eventually, we show with some examples that uniqueness cannot be expected in general in the geometric case.

“Under a monotonicity assumption we can also prove uniqueness of the limit solution, even though this remains an open question in full generality.”

This packet proves two results toward that question. First, a relative Bregman estimate gives uniqueness for genuinely nonlinear uniformly convex integrands whenever one of the two solutions has an integrable L_x^∞ bound on $D\dot{v}$. Second, the nonlinear error vanishes identically for every constant-coefficient strongly elliptic quadratic integrand. This yields unconditional uniqueness, contraction, and whole-family minimizing-movement convergence for that complete anisotropic quadratic class.

2 Framework and definitions

Let $\Omega \subset \mathbb{R}^d$ be bounded, connected, and Lipschitz. Put

$$\Psi(w) = \frac{1}{2} \|w\|_{L^1(\Omega)}^2, \quad \mathcal{J}_1 = \partial\Psi \subset L^1(\Omega) \times L^\infty(\Omega).$$

For $w \neq 0$, the relation $q \in \mathcal{J}_1(w)$ is equivalent to

$$\|q\|_\infty = \|w\|_1, \quad qw = \|w\|_1 |w| \quad \text{a.e.}, \quad (1)$$

while $\mathcal{J}_1(0) = \{0\}$. In particular, $\mathcal{J}_1$ is monotone:

$$\int_\Omega (q_1 - q_2)(w_1 - w_2) dx \geq 0 \quad (q_i \in \mathcal{J}_1(w_i)). \quad (2)$$

Write $A = \nabla F$ and $H = D^2 F$. A sufficiently regular solution of the flow characterized in [1, Theorem 2.15] satisfies

$$q_u := \operatorname{div} A(Du) \in \mathcal{J}_1(\dot{u}) \quad \text{for a.e. time.} \quad (3)$$

The pointwise formulation in the source paper,

$$|q_u| \leq \|\dot{u}\|_1, \quad \dot{u} q_u = |\dot{u}| \|\dot{u}\|_1,$$

is precisely (3). We use either common Dirichlet data, so that $u - v$ and $\dot{u} - \dot{v}$ have zero trace, or the natural Neumann condition $A(Du) \cdot n = A(Dv) \cdot n = 0$.

For $a, b \in \mathbb{R}^d$, define the one-sided Bregman divergence

$$\mathcal{B}_F(a \mid b) = F(a) - F(b) - A(b) \cdot (a - b)$$

and the relative energy

$$\mathcal{E}_F(u \mid v) = \int_\Omega \mathcal{B}_F(Du \mid Dv) dx.$$

3 Proof intuition

Monotonicity of the L^1 duality map already gives the favorable term

$$\int_\Omega [A(Du) - A(Dv)] \cdot D(\dot{u} - \dot{v}) dx \leq 0.$$

For a quadratic F , this is exactly the time derivative of the energy of $D(u - v)$, which is the contraction used in the source paper. For nonlinear F , differentiating the relative Bregman energy produces one additional Taylor remainder:

$$[A(Du) - A(Dv) - H(Dv)(Du - Dv)] \cdot D\dot{v}.$$

If H is Lipschitz, this is quadratic in $Du - Dv$. Uniform convexity says that the Bregman energy is also quadratic in that difference. Thus an integrable L^∞ bound on $D\dot{v}$ closes a Gronwall inequality.

This mechanism is asymmetric on purpose: it is a weak–strong estimate. It also identifies the exact obstruction to using the quadratic proof in the full weak class. The regularity obtained in [1] controls $D\dot{v}$ in L^2 , not in $L_t^1 L_x^\infty$, so the theorem below does not silently resolve the remaining open case.

4 The relative-Bregman estimate

Theorem 1 (Nonlinear weak–strong uniqueness). *Assume $F \in C^2(\mathbb{R}^d)$ and that there are constants $0 < \gamma \leq \Gamma < \infty$ and $K < \infty$ such that*

$$\gamma I \leq H(\xi) \leq \Gamma I \quad (\xi \in \mathbb{R}^d), \quad (4)$$

$$\|H(\xi) - H(\eta)\| \leq K|\xi - \eta| \quad (\xi, \eta \in \mathbb{R}^d). \quad (5)$$

Let $u, v \in AC([0, T]; H^1(\Omega))$ satisfy (3), the same boundary condition, and suppose

$$D\dot{v} \in L^1(0, T; L^\infty(\Omega)).$$

Assume the integrations by parts in (9) below are valid (for example, $q_u, q_v \in L^1(0, T; L^\infty)$ with the stated boundary conditions). Then for $0 \leq s \leq t \leq T$,

$$\mathcal{E}_F(u(t) \mid v(t)) \leq \mathcal{E}_F(u(s) \mid v(s)) \exp\left(\frac{K}{\gamma} \int_s^t \|D\dot{v}(r)\|_\infty dr\right). \quad (6)$$

Consequently, two such solutions with the same initial datum coincide.

Proof. Set $a = Du$ and $b = Dv$. The hypotheses on H imply the quadratic equivalence

$$\frac{\gamma}{2}|a - b|^2 \leq \mathcal{B}_F(a \mid b) \leq \frac{\Gamma}{2}|a - b|^2. \quad (7)$$

The Banach-space chain rule (or a direct calculation for smooth curves, followed by approximation) gives

$$\begin{aligned} \frac{d}{dt} \mathcal{B}_F(a \mid b) &= [A(a) - A(b)] \cdot (\dot{a} - \dot{b}) \\ &\quad + [A(a) - A(b) - H(b)(a - b)] \cdot \dot{b}. \end{aligned} \quad (8)$$

After integration over Ω , the first term is

$$\begin{aligned} P(t) &:= \int_\Omega [A(Du) - A(Dv)] \cdot D(\dot{u} - \dot{v}) dx \\ &= - \int_\Omega (q_u - q_v)(\dot{u} - \dot{v}) dx \leq 0. \end{aligned} \quad (9)$$

Here the equality is integration by parts, and the inequality is the monotonicity (2).

The integral form of Taylor's theorem and (5) yield

$$|A(a) - A(b) - H(b)(a - b)| \leq \frac{K}{2}|a - b|^2. \quad (10)$$

Combining (8)–(10) and using (7), we obtain for a.e. t ,

$$\frac{d}{dt} \mathcal{E}_F(u \mid v) \leq \frac{K}{2} \|D\dot{v}\|_\infty \int_\Omega |Du - Dv|^2 dx \leq \frac{K}{\gamma} \|D\dot{v}\|_\infty \mathcal{E}_F(u \mid v).$$

Gronwall's inequality proves (6). If $u(0) = v(0)$, then (7) gives $Du(t) = Dv(t)$ for every t .

With Dirichlet data, $u = v$ follows immediately. With Neumann data, $u(t) - v(t)$ is spatially constant. Lemma 2 below shows that its time derivative is zero; the initial equality then gives $u = v$. $\square$

Lemma 2 (Neumann constant rigidity). *Let Ω be connected. If $f \in H^1(\Omega)$, $c \in \mathbb{R}$, and $q \in L^\infty(\Omega)$ satisfy*

$$\int_\Omega q dx = 0, \quad q \in \mathcal{J}_1(f) \cap \mathcal{J}_1(f + c),$$

then $c = 0$.

Proof. If $q = 0$, then (1) gives $f = f + c = 0$. Suppose $q \neq 0$. Since q has mean zero, the sets $\{q > 0\}$ and $\{q < 0\}$ have positive measure. Relation (1) gives

$$f \geq 0 \text{ on } \{q > 0\}, \quad f + c \leq 0 \text{ on } \{q < 0\}.$$

If $c > 0$, these say that f lies in $[0, \infty)$ on one positive-measure set and in $(-\infty, -c]$ on another. Moreover, on $\{q = 0\}$, (1) for the two nonzero vectors forces both $f = 0$ and $f + c = 0$; hence this set is null if $c \neq 0$. A Lipschitz truncation of f mapping $(-\infty, -c]$ to 0 and $[0, \infty)$ to 1 is therefore an H^1 function taking only the values 0 and 1 a.e., with both level sets of positive measure. Such an H^1 function has zero weak gradient and must be constant on connected Ω , a contradiction. The case $c < 0$ is symmetric. $\square$

Corollary 3 (A genuinely nonlinear family). *For every $\varepsilon > 0$, the integrand*

$$F_\varepsilon(\xi) = \frac{1}{2} |\xi|^2 + \varepsilon \sum_{j=1}^d \log \cosh(\xi_j)$$

satisfies the hypotheses of Theorem 1. Thus its flow is unique in the class of that theorem.

Proof. Its Hessian is diagonal with entries $1 + \varepsilon \operatorname{sech}^2(\xi_j)$. Hence $I \leq H \leq (1 + \varepsilon)I$, and the derivative of the Hessian is globally bounded. Theorem 1 applies. $\square$

5 The full strongly elliptic quadratic subcase

Corollary 4 (Anisotropic quadratic contraction and uniqueness). *Let M be a constant symmetric positive-definite $d \times d$ matrix and*

$$F(\xi) = \frac{1}{2} \xi \cdot M \xi.$$

For any two energy-class solutions of (3) with common Dirichlet data or homogeneous Neumann data,

$$\frac{1}{2} \int_\Omega D(u(t) - v(t)) \cdot MD(u(t) - v(t)) dx \leq \frac{1}{2} \int_\Omega D(u(s) - v(s)) \cdot MD(u(s) - v(s)) dx. \quad (11)$$

No L^∞ assumption on $D\dot{v}$ is needed. In particular the flow from a fixed initial datum is unique.

Moreover, the unique discrete minimizers of

$$\Phi(w) + \frac{1}{2\tau} \|w - w^{n-1}\|_1^2, \quad \Phi(w) = \frac{1}{2} \int_{\Omega} Dw \cdot MDw,$$

converge, without choosing a subsequence, to this flow as $\tau \downarrow 0$ in the same topologies as the quadratic minimizing movements in [1].

Proof. Here $H \equiv M$, so $K = 0$ and the Taylor remainder in (8) vanishes identically. Therefore

$$\frac{d}{dt} \frac{1}{2} \int_{\Omega} D(u - v) \cdot MD(u - v) dx = - \int_{\Omega} (q_u - q_v)(\dot{u} - \dot{v}) dx \leq 0,$$

which proves (11). Dirichlet uniqueness is immediate; Neumann uniqueness follows from Lemma 2.

The existence, time-regularity, and compactness estimates in Section 4 of [1] use only polarization of the quadratic form, ellipticity, and the monotonicity of $\mathcal{J}_1$. Replacing $|D \cdot|^2$ by $D \cdot MD \cdot$ changes the estimates only by the two ellipticity constants of M . Thus every vanishing-step sequence has a subsequence converging to an energy-class solution. Uniqueness shows that all cluster points agree. Relative compactness then upgrades subsequential convergence to convergence of the whole family. $\square$

6 Verification notes and limitations

1. **Sign check.** The source equation has $q = \operatorname{div} \nabla F(Du) \in \mathcal{J}_1(\dot{u})$. Hence monotonicity of $\mathcal{J}_1$ makes $-\int (q_u - q_v)(\dot{u} - \dot{v})$ nonpositive, exactly as required.
2. **Algebra check.** The script `code/check_bregman_identity.py` numerically differentiates the relative Bregman energy for the nonlinear integrand in Corollary 3 and compares it with the two terms in (8). It also tests the Taylor remainder bound on 10,000 seeded random inputs. This is a regression/sign check, not a proof.
3. **What remains open.** The source paper proves only $\dot{u} \in L_{\operatorname{loc}}^2((0, \infty); H^1(\Omega))$ for its general strongly convex limit solutions. That does not imply $D\dot{u} \in L_t^1 L_x^\infty$. Theorem 1 therefore gives a complete classical/weak-strong uniqueness criterion but not unconditional uniqueness for every source-paper limit solution.
4. **Neumann issue.** Relative energy controls gradients only. Lemma 2 is essential; it rules out a time-dependent spatial constant by using the L^1 duality relation and connectedness.
5. **Novelty status.** A bounded search on 22 July 2026 used the exact title, arXiv:2302.12786, DOI 10.1137/22M1527556, the quoted open phrase, and combinations of “ L^1 gradient flow”, “strongly convex”, and “uniqueness”. Sources checked included arXiv, the 2024 SIAM journal page, CVGMT, and general web/arXiv search. The journal version still states the full question as open, and no later paper explicitly closing it was found. The relative-Bregman argument may overlap standard weak-strong relative entropy techniques, so novelty confidence for the method is moderate; the mathematical claim is presented as a candidate partial result, not as a priority claim.

7 Human-review recommendation

The packet is recommended for expert review as a likely valid partial result. The key checks are: (i) the chain-rule regularity in the desired solution class; (ii) the Neumann constant-rigidity lemma; and (iii) that the source paper’s Section 4 estimates transfer to a constant SPD matrix exactly as claimed. A useful next push would prove smoothing strong enough to place every positive-time limit flow in $L_t^1 W_x^{1,\infty}$, or replace the L^∞ factor in the commutator estimate by a norm already controlled by the source paper.

References

- [1] A. Chambolle and M. Novaga, *L^1 -gradient flow of convex functionals*, SIAM J. Math. Anal. **56** (2024), no. 5, 5747–5781; arXiv:2302.12786; DOI: 10.1137/22M1527556.